Software Requirements Specification (SRS)

Project: Real-Time Collaborative Virtual Whiteboard

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Status: Final for submission — team sign-off pending (see Approvals); written before implementation (historical); implemented and tested in the execution of 12-10-2026 (uncommitted working tree on base commit 82af885): 47 of 50 requirements verified (RTM status P); NFR-006 and NFR-007 (TC-021) and SEC-REQ-007 (TC-109) not verified because those tests are Blocked; see `docs/IMPLEMENTATION_STATUS.md`. Update 17-10-2026: TC-021 is Pass with limitations (STP TC-021 record), so 49 of 50 requirements are verified (RTM status P); only SEC-REQ-007 is not verified (TC-109 Blocked). Update 02-11-2026: final validation of `main` after PR #33 (`docs/reports/RELEASE_READINESS_2026-11-02.md`); RTM statuses unchanged; two implemented functions outside the requirement baseline (PNG export, board rename) are recorded in §2.7 pending a scope decision.

Revision history

Version	Date	Author	Change summary	Approval
0.1	10-10-2026	Rachit, Prithvi (draft)	First complete draft for team review	Pending
0.2	10-10-2026	Rachit, Prithvi (draft)	Version-history rules made consistent (D5b); FR-008 board size limit added; NFR measurement definitions; guest-token expiry; ownership-transfer details; FR-049 test moved to TC-023; RTM status rules	Pending
0.3	10-10-2026	Rachit, Prithvi (draft)	FR-008 limit defined on BSON size (15 MiB); FR-007 clear rejected if the pre-clear version cannot be stored; permission matrix lists members view; measurable acceptance exit criterion; numbered figures; RTM columns as in the template	Pending
0.4	10-10-2026	Rachit, Prithvi (draft)	FR-008 rewritten with explicit size measures and boundary/deletion/version acceptance criteria; message limit stated as 65,536 bytes per packet (SEC-REQ-005, §3.4); FR-007 and FR-033 cover version-write failure and interrupted-restore recovery (TC-024)	Pending
1.0	11-10-2026	Rachit, Prithvi	Final for submission: interrupted-clear and restore recovery and idempotent restore in FR-007/FR-033 (TC-024, TC-025); design values stated with decision references; NFR targets specified (verification pending); SEC-REQ-008 log fields and `null` representation defined; FR-033 distinguishes new restores, retries and key reuse; SEC-REQ-006 login limit per entered e-mail; SEC-REQ-008 lists the affected account (`subjectId`)	Team sign-off pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
SRS Lead	Rachit S Mane		
SAD Lead	P Chinni Krishna Reddy		
STP Lead	Pranav V Menon		
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional requirements, non-functional requirements, security requirements, external interfaces and acceptance criteria of the Real-Time Collaborative Virtual Whiteboard developed by Team 11. It is the baseline against which the Software Architecture and Design Specification (SAD) and the Software Test Plan (STP) are written. Every requirement has a stable identifier that is reused unchanged in the SAD, the STP and the Requirements Traceability Matrix (RTM).

1.2 Scope

The system is a web application that allows several people to draw, write and edit the same whiteboard at the same time from their own browsers. Changes made by one participant are synchronised to every other participant of the board in real time, and the system keeps a basic version history from which the board owner can restore an earlier state.

In scope for version 1.0:

- Registered user accounts, and guest access to a board through a shareable link.
- Owner, Editor and Viewer roles with server-side enforcement.
- Freehand drawing, erasing, basic shapes, text, colour and stroke-width controls, and basic object editing.

- Real-time synchronisation with server-authoritative ordering and recovery after reconnection.
- Participant presence on a board.
- Automatic version history with read-only preview and owner-only restore.

Out of scope for version 1.0 (candidate enhancements, see §2.2): undo/redo across users, live cursors, PNG/PDF export, chat or comments, advanced version comparison, mobile-specific optimisation, offline editing, automatic resubmission of unacknowledged changes after a disconnection, chunked storage of very large boards, and deployment on multiple server instances.

1.3 Audience

Team 11 developers and testers, the course instructor and teaching assistants who assess the project, and any reviewer who needs to verify the system against its requirements.

1.4 Definitions and acronyms

Term	Meaning
Board	A single shared whiteboard with its own content, members, share links and version history.
Board object	One item on a board: a freehand stroke, a shape (rectangle, ellipse, line) or a text object.
Operation	A single change submitted to the server: add object, update object, delete object or clear board.
Participant	A user (registered or guest) currently connected to a board.
Registered user	A person with an account (e-mail address and password).
Guest	A person who joins a board through a share link without logging in.
Member	A registered user who has a stored role on a board.
Owner / Editor / Viewer	Board roles; permissions are defined in §4.4 (FR-051).
Share link	A URL containing a secret token that grants the Viewer or Editor role on one board until it expires or is revoked.
Version	A saved, numbered snapshot of the complete content of a board.
Changed content	The state of a board whose current objects differ from its latest version (§4.3, rule 1).
T	The periodic version interval, 5 minutes (§4.3, rule 4).
Restore	Making the content of an earlier version the current board content by creating a new version.
Sequence number	A per-board number assigned by the server to every accepted operation; it defines the order of operations.
TB-STD	The standard test board defined in DECISIONS.md D7a and STP §6, used to measure performance targets.
AC	Acceptance criterion.
API	Application Programming Interface.
LWW	Last write wins.
REST	Representational State Transfer (HTTP-based API style).
RTM	Requirements Traceability Matrix.
SAD	Software Architecture and Design Specification.
STP	Software Test Plan.
TLS	Transport Layer Security.

Term	Meaning
UI	User interface.
XSS	Cross-site scripting.

1.5 References

- Team 11 Software Architecture and Design Specification (SAD), version 1.0.
- Team 11 Software Test Plan (STP), version 1.0.
- Team 11 Decision Log (`docs/DECISIONS.md`) and ID registry (`docs/traceability/id-registry.md`).
- Software Engineering Mini-Project Deliverables Part-1 (course brief) and the supplied SRS template.

2. Overall description

2.1 Product perspective

The whiteboard is a new, self-contained web application. It consists of a browser client used by participants, a server that authenticates users, enforces permissions, orders and broadcasts board operations and manages version history, and a database that stores accounts, boards, share links, board content and versions. It has no dependency on external business systems. The selected implementation technologies are described in the SAD (§3.6); they are design decisions, not requirements of this SRS.

2.2 Major product functions

Core functions (mandatory for version 1.0):

- Register, log in and log out.
- Create, open, list and delete boards.
- Share a board through Viewer or Editor links that expire and can be revoked.
- Manage board members and roles; transfer board ownership.
- Draw freehand strokes, add rectangles, ellipses, lines and text, select colour and stroke width.
- Select, move, restyle and delete objects; erase objects; clear the whole board (Owner only).
- Synchronise every accepted change to all participants in real time, with consistent ordering.
- Show which participants are currently on the board.
- Recover the current board state after a lost connection.
- Create versions automatically; list and preview versions; restore a version (Owner only).
- Reject unauthorised or invalid requests on the server.

Candidate enhancements (not part of version 1.0; may be added only through a new, approved requirement): undo/redo across users, live cursors and activity indicators,

PNG/PDF export, chat or comments, advanced version comparison, mobile-specific optimisation.

The official problem statement for Team 11 requires simultaneous drawing, writing and editing, real-time synchronisation and basic version history; all three are core functions above.

2.3 User roles and characteristics

Role	Description	Characteristics
Visitor	Anyone who has not logged in and is not on a board.	Can register, log in, or open a share link.
Registered user	Has an account.	Can create boards and becomes their Owner; can join boards through links.
Guest	Joins one board through a share link without an account.	Access lasts only for the browser session and only while the link is valid.
Owner	The registered user who owns a board (exactly one per board).	Full control of content, sharing, members, versions and deletion.
Editor	Member or guest with edit rights on a board.	Can create and change content.
Viewer	Member or guest with read-only rights on a board.	Can see the board, participants and version history.

All users are expected to be familiar with a web browser and a mouse or touchpad. No training is assumed.

2.4 Operating environment

- Client: current stable desktop versions of Google Chrome, Mozilla Firefox and Microsoft Edge with a browser viewport at least 1280 px wide (NFR-006).
- Server: a single server instance hosting the application server and the database (development and test environment: one laptop).
- Network: participants reach the server over a local network or the Internet; deployed systems use encrypted transport (SEC-REQ-007).

2.5 Design and implementation constraints

- The selected technology stack is MERN with Socket.IO (DECISIONS.md D10, SAD §3.6).
- Version 1.0 runs as a single server instance; horizontal scaling is out of scope.
- The content of one board is limited to 15 MiB measured as BSON-encoded size (FR-008) so that every version snapshot fits in one MongoDB document (16 MiB BSON limit).
- The product must be developed and tested by a team of four students within the course schedule, using a shared GitHub repository with protected `main`, feature branches, Pull Requests and CI.
- No real personal data beyond an e-mail address and display name is collected.

2.6 Assumptions and dependencies

- Participants have a stable network connection for most of a session; short interruptions are handled by FR-025.

- All participants of a board use a supported browser (NFR-006).
- Performance targets (NFR-001 to NFR-005) are to be verified in the local test environment defined in STP §6, not on the public Internet; they were measured there in the execution of 12-10-2026 (uncommitted working tree on base commit 82af885; see `docs/IMPLEMENTATION_STATUS.md`) and met (historical v1.0 note: not measured at the time of writing).
- The design decisions recorded with their rationale in `docs/DECISIONS.md` are confirmed by all four members when they sign this document.

2.7 Open items

The requirements in this document are complete for version 1.0. The following items depend on actions outside the document and are tracked in `docs/DECISIONS.md` (E1–E5):

- Sign-off of this SRS by all four team members (approval table).
- Confirmation of the current submission deadline with the instructor (the brief's ETA of 28-09-2026 has passed).
- Measured verification of NFR-001 to NFR-005: done in the execution of 12-10-2026 (TC-009, TC-013, TC-019, TC-020 Pass; local hardware only, see STP Appendix A).
- Verification of NFR-006 and NFR-007: done on 17-10-2026 (TC-021 Pass with limitations: manual in Firefox 158 and Chrome, automated in Chrome and Edge; limitations recorded in the STP TC-021 record).
- A deployed HTTPS environment for verifying SEC-REQ-007 (TC-109).
- **Scope decision pending (02-11-2026): functions implemented outside the version 1.0 requirement baseline.** The requirements of this SRS are unchanged; the following functions were implemented during the sprints (GitHub issues of the team's sprint plan) without a new SRS requirement, ID or RTM entry:
  - *PNG export* (issue #16, PR #28): any participant can download the board as a PNG image. §1.2 and §2.2 list "PNG/PDF export" as a candidate enhancement that "may be added only through a new, approved requirement"; no such approval is recorded. The function is tested by automated tests outside the STP test-case set (`client/tests/exportImage.test.js`, three E2E tests in `e2e/boards.spec.cjs`).
  - *Board rename* (issue #13, PR #28): the Owner can rename a board from the dashboard (`PATCH /api/v1/boards/{boardId}`, SAD API-02). FR-042 to FR-044 cover create, list and delete only; the FR-051 permission matrix has no rename row (implemented as Owner-only). Tested in `server/tests/integration/boards.test.js` and `e2e/boards.spec.cjs`.

The team and the course mentor must decide either to bring these functions into version 1.0 (new requirement IDs allocated in `docs/traceability/id-registry.md`, test cases and RTM rows) or to record them as accepted additions outside the version 1.0 baseline. Until then they are not counted in the requirement coverage.

3. External interface requirements

3.1 User interfaces

- **Board view:** a canvas occupying most of the window; a toolbar with pen, eraser, select, rectangle, ellipse, line and text tools and colour and stroke-width controls; a participant panel; a version-history panel; the user's role and the connection status always visible (NFR-007).
- **Role-aware controls:** tools that the user's role does not permit are hidden or disabled (e.g. drawing tools for Viewers; Clear board, Share and Restore for non-Owners). Disabled controls are a convenience only; enforcement is on the server (SEC-REQ-003).
- **Dashboard:** list of the user's boards with title, role and last-modified time; actions to create, open and (for Owners) delete a board.
- **Sharing dialog (Owner):** create a link by choosing a role and an expiry; list active links with role and expiry; revoke a link.
- **Members dialog (Owner):** list members with roles; change role, remove member, transfer ownership to an Editor or Viewer member after a confirmation dialog (FR-049, FR-050).
- **Guest entry page:** shown when a share link is opened without login; asks for a display name or offers login.
- **Messages:** confirmation dialogs for destructive actions (clear board, delete board, restore version, revoke link, transfer ownership); clear error messages for rejected actions, a full board (FR-008), expired or revoked links, and lost connection.
- **Accessibility:** all toolbar controls have text labels or tooltips and can be reached and activated with the keyboard (NFR-007).
- **Viewport:** the board view is usable at a browser viewport width of 1280 px or more without horizontally cut-off controls (NFR-006).

3.2 Hardware interfaces

No special hardware is required. The client uses the standard pointer device (mouse, touchpad or pen tablet acting as a mouse) and keyboard of the user's computer.

3.3 Software interfaces

- **Web browser:** the client runs in a supported browser (NFR-006) and uses its HTML canvas, WebSocket and HTTP capabilities.
- **Client–server request interface:** an HTTP request/response interface for accounts, boards, members, share links and version history (specified in SAD §4.3, API-02 and API-03).
- **Client–server real-time interface:** a persistent bidirectional connection for board operations, presence and restore notifications (specified in SAD §4.3, API-01).
- **Database:** the server stores all persistent data in a database (MongoDB, SAD §3.6). No other external software system is integrated.

3.4 Communications interfaces

- All client–server traffic in a deployed system uses HTTPS and secure WebSockets (TLS 1.2 or later) (SEC-REQ-007).
- Every request and real-time connection carries either a session token (registered users) or a share-link-derived guest token (guests) (SEC-REQ-002).
- Real-time messages are JSON objects; the maximum size of one message is 64 KiB (65,536 bytes per Socket.IO packet) (SEC-REQ-005). This limit is separate from the board content limit of FR-008.
- When the real-time connection drops, the client retries automatically and resynchronises on reconnection (FR-025).

4. System features (detailed)

Each requirement below has a stable identifier, an acceptance criterion (AC-<Req ID>) and a reference to the STP test case(s) that verify it. Identifiers are allocated in blocks: FR-001–019 canvas editing, FR-020–029 real-time synchronisation, FR-030–039 version history, FR-040–059 accounts, boards, sharing and access. Identifiers are never renumbered; a removed requirement is marked "Withdrawn" in `docs/traceability/id-registry.md`.

"Editing roles" means the Owner and Editor roles. All operations in §4.1 are subject to the permission matrix in FR-051 and to server-side validation (SEC-REQ-003, SEC-REQ-005).

4.1 Canvas editing

Description: participants with an editing role create and change board objects; every accepted change is synchronised to all participants (§4.2).

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-001	The system shall let a user with an editing role draw a freehand stroke on the board with the pen tool, using the currently selected colour and stroke width.	Functional	High	Problem statement	AC-FR-001: After a pointer drag with the pen tool, one stroke object following the drag path, in the selected colour and width, appears on the board. Test: TC-001	FR-002, FR-020
FR-002	The system shall let a user with an editing role select a stroke colour (any `#RRGGBB` value, with a palette of preset colours) and a stroke width from 1 to 20 px, and apply them to new strokes and shapes.	Functional	High	Team	AC-FR-002: Objects created after a change of colour or width use the new values; the width control does not accept values outside 1–20 px. Test: TC-001	Values: DECISIONS.md D1a
FR-003	The system shall let a user with an editing role add a rectangle, an ellipse or a straight line by dragging on the board with the corresponding tool.	Functional	High	Problem statement	AC-FR-003: Each of the three tools creates one object of the matching type whose bounding box or end points equal the drag start and end points. Test: TC-002	FR-002
FR-004	The system shall let a user with an editing role add a text object at a chosen position and edit its text, up to 500 characters per text object.	Functional	High	Problem statement ("write")	AC-FR-004: Typed text appears at the chosen position; input beyond 500 characters is not accepted; edited text replaces the previous text for all participants. Test: TC-002	Text is displayed as plain text (SEC-REQ-005)
FR-005	The system shall let a user with an editing role select one object and move it, change its colour or stroke width, or delete it.	Functional	High	Problem statement ("edit")	AC-FR-005: After each action the object shows the new position or style, or is removed, on every participant's board. Test: TC-003	FR-020, FR-023
FR-006	The system shall provide an eraser tool that deletes every whole object the eraser path touches.	Functional	High	Team	AC-FR-006: Objects touched by the eraser path are removed completely; objects not touched remain unchanged. Test: TC-004	Partial (pixel) erasing is not supported

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-007	The system shall let only the Owner clear the entire board after an explicit confirmation; before clearing, the system shall apply the "before clear" version trigger of FR-030 so that the pre-clear content is preserved in the latest version; if that version cannot be stored, the clear shall be rejected. Clearing an empty board shall have no effect.	Functional	High	Team	AC-FR-007: After the Owner confirms, the board is empty for all participants and the latest version contains the pre-clear content (a new "before clear" version if the content had changed since the latest version, otherwise the existing latest version); clearing an empty board changes nothing and creates no version; if the pre-clear version cannot be stored, the clear is rejected and the board keeps its content; a clear interrupted by a server failure after it was committed is completed when the server restarts, and no participant receives a success acknowledgement before it is complete; a clear request from an Editor or Viewer is rejected. Test: TC-005, TC-025	FR-030 (rules in §4.3), FR-051
FR-008	The system shall limit the **content size** of one board to 15 MiB (15,728,640 bytes) and shall reject, with a "board full" message, any add or update operation that would make the content size exceed the limit; a rejected operation shall not be stored, acknowledged as accepted or broadcast. The content size is the BSON-based count of the board's object list defined in SAD §3.6 (measure M3); it is independent of the per-message size limit of SEC-REQ-005 (measure M1).	Functional	Medium	Team (D11c)	AC-FR-008: (a) an operation after which the content size is at most 15,728,640 bytes is accepted; (b) an operation that would make it exceed 15,728,640 bytes by one byte or more is rejected with "board full", is not stored, not acknowledged as accepted and not broadcast, and the board is unchanged; (c) after objects are deleted, an operation that fits within the limit again is accepted; (d) a version of a board within one object of the limit is stored, and the stored version document is below MongoDB's 16 MiB (16,777,216-byte) document limit. Test: TC-022	Keeps every version snapshot below the 16 MiB document limit (SAD §3.6); value and accounting per D11c

4.2 Real-time synchronisation

Description: the server is the single authority on board state. It accepts or rejects each operation, gives every accepted operation a per-board sequence number and sends it to all participants, who apply operations in sequence-number order. An update carries

the complete new value of each property it changes. This policy (DECISIONS.md D6) has a known limitation: when two users change the same property of the same object at nearly the same time, the change accepted first by the server is overwritten by the later one and is not merged; changes to different properties of the same object are both kept.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-020	The system shall deliver every operation accepted by the server to all other participants of the same board without the participants reloading the page.	Functional	High	Problem statement	AC-FR-020: An operation made by one participant appears on every other connected participant's board without any user action; timing is verified under NFR-001. Test: TC-001, TC-002	NFR-001
FR-021	The server shall assign each accepted operation a sequence number that is unique and strictly increasing within the board, and every client shall apply operations in sequence-number order.	Functional	High	Team (D6)	AC-FR-021: In the operation logs of all clients for a test session, the sequence numbers are strictly increasing with no duplicates, and every client applies them in the same order. Test: TC-006, TC-007	Basis for FR-022, FR-023
FR-022	The system shall preserve concurrent operations on different objects independently, so that no accepted operation on one object changes or removes another object.	Functional	High	Team (D6)	AC-FR-022: When two participants change two different objects at the same time, both changes are present on all boards afterwards. Test: TC-006	—
FR-023	When two or more accepted operations change the same property of the same object, the system shall make the value from the operation with the highest sequence number the final value of that property (last write wins); an update to an object that has already been deleted shall be rejected.	Functional	High	Team (D6)	AC-FR-023: After concurrent updates of the same property of one object, every participant shows the value of the update with the higher sequence number; after concurrent updates of different properties, both values are present; an update arriving after the object's deletion is rejected and the object stays deleted. Test: TC-007	Limitation documented in §4.2 description
FR-024	The system shall send a participant the complete current board content, the current sequence number and the participant list when the participant opens or joins a board.	Functional	High	Team	AC-FR-024: A participant who joins after other participants have made changes sees exactly the same objects as existing participants. Test: TC-008	FR-052

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-025	When the real-time connection is lost, the client shall show a "disconnected" indicator, disable editing tools and retry the connection automatically; after reconnecting it shall replace its board content with the current server state before re-enabling editing. Local changes that had not been acknowledged by the server before the connection was lost shall be discarded, and the user shall be told that recent unconfirmed changes may not have been saved.	Functional	High	Team (D6a)	AC-FR-025: During the outage the indicator is shown and drawing is not possible; after connectivity returns, the board matches the server state (including changes made by others during the outage) within the time in NFR-004; if unacknowledged changes existed, the notice is shown. Test: TC-009	NFR-004; offline editing and automatic resubmission are out of scope
FR-026	When the server rejects an operation, the system shall inform the submitting participant with a message stating the reason and shall remove any local, unconfirmed effect of that operation from the participant's board.	Functional	High	Team	AC-FR-026: Within 1 second of the rejection reaching the client, the submitter sees an error message and the rejected change is no longer visible on the submitter's board; no other participant receives it. Test: TC-010	SEC-REQ-003, SEC-REQ-005
FR-027	The system shall store every accepted operation's effect persistently before acknowledging or broadcasting it, so that the board content survives all participants leaving and a server restart.	Functional	High	Team	AC-FR-027: After all participants leave and the server is restarted, reopening the board shows every change that had been acknowledged before the restart. Test: TC-011	—

4.3 Version history

Description: basic version history is a mandatory core feature (official problem statement). The system saves numbered versions of the board automatically. All roles can view the history; only the Owner can restore. Restoring never deletes history: it creates a new version whose content is a copy of the selected version. Storage format and the retention mechanism are specified in the SAD (§3.6).

Version rules (DECISIONS.md D5b):

1. **Changed content.** The board has *changed content* when its current objects differ from the objects of its latest version (or when it has objects and no version exists yet).
2. **No-change rule.** An automatic trigger — periodic, before clear, last participant left — creates a version only if the board has changed content at that moment. Otherwise no version is created, because the latest version already holds identical content.
3. **Clear.** Before the board is cleared, the system applies the "before clear" trigger, so the pre-clear content is always preserved — in a new version if it had changed, or in the existing latest version if it had not. Clearing an already empty board has no effect and creates no version.
4. **Periodic interval.** The interval *T* is 5 minutes (DECISIONS.md D11a), set through a configuration parameter; tests record the value configured in the build under test.
5. **Restore.** A restore always creates a new version (trigger "restore"), even if the restored content equals the current content, so every restore is recorded. It is the only exception to the no-change rule.
6. **Retention.** At least the 20 most recent versions are kept (FR-035). Older versions may be removed by the retention mechanism; nothing else ever modifies or deletes a version.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-030	The system shall create a version of a board automatically, subject to the no-change rule, (a) every *T* minutes while the board has changed content, (b) before the board is cleared, and (c) when the last participant leaves the board.	Functional	High	Problem statement	AC-FR-030: Each trigger creates exactly one version with the board content at that moment when the board has changed content, and no version when it has not; after a clear, the latest version contains the pre-clear content. Test: TC-005, TC-012	*T* = 5 min (D11a); rules in §4.3 (D5b)
FR-031	The system shall show every participant of a board the list of its retained versions, newest first, with version number, creation date and time, and trigger (periodic, before clear, last participant left, or restore of version *n*).	Functional	High	Team	AC-FR-031: For Owner, Editor and Viewer the version panel lists all retained versions in descending version-number order with the four listed fields. Test: TC-012	FR-051

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-032	The system shall let any participant preview a selected version read-only without changing the live board for anyone.	Functional	Medium	Team	AC-FR-032: The preview shows exactly the selected version's objects; no editing is possible in the preview; the live board of all participants, including changes made during the preview, is unaffected. Test: TC-012	—
FR-033	The system shall let only the Owner restore a selected retained version, after confirmation, by creating a new version whose content equals the selected version and making that content the current board content; a restore shall not modify or delete any existing version.	Functional	High	Problem statement, Team (D5)	AC-FR-033: After restoring version *n*, the version list contains all previously retained versions unchanged plus one new version "restore of *n*", and the board content equals version *n*; a restore request from an Editor or Viewer is rejected and creates no version; a restore interrupted by a server failure after it was committed is completed when the server restarts, so that the board content equals version *n* and exactly one "restore of *n*" version exists; a restore whose new version would exceed the storage limit, or that fails before it was committed, is rejected and leaves the board and the version list unchanged; a new restore request (new idempotency key) always creates a new version; a retried request with the same idempotency key and version returns the recorded result (same new version number and sequence number) without restoring again or creating a version; the same key used for a different version is rejected; concurrent restores of one board are applied one at a time, each with its own version. Test: TC-013, TC-024, TC-102	FR-051; retention per FR-035

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-034	After a restore, the system shall replace the board content of every connected participant with the restored content.	Functional	High	Team (D5)	AC-FR-034: All connected participants show the restored content without reloading; timing is verified under NFR-005. Test: TC-013	NFR-005
FR-035	The system shall retain at least the 20 most recent versions of each board until the board is deleted.	Functional	Medium	Team (D5a)	AC-FR-035: After 25 versions have been created on a board, at least the 20 most recent are listed and can be previewed and restored; a request for a version that is no longer retained returns "version not found". Test: TC-014	Value per D5a; mechanism in SAD §3.6

4.4 Accounts, boards, sharing and access

Description: registered users own boards and share them through links that grant the Viewer or Editor role. Every action is checked against the permission matrix below on the server (SEC-REQ-003), for both request/response calls and real-time messages.

Permission matrix (FR-051)

Action	Owner	Editor	Viewer
View board, participants, members list and version list; preview a version	Yes	Yes	Yes
Draw, add shapes and text, edit, move, restyle, erase, delete objects	Yes	Yes	No
Clear the entire board	Yes	No	No
Restore a version	Yes	No	No
Create, list and revoke share links	Yes	No	No
Change member roles, remove members	Yes	No	No
Transfer ownership, delete the board	Yes	No	No

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-040	The system shall let a visitor register an account with a unique e-mail address, a password that satisfies SEC-REQ-001 and a display name of 1–30 characters.	Functional	High	Team (D2)	AC-FR-040: Valid data creates an account; a duplicate e-mail address or invalid data is rejected with a message naming the problem. Test: TC-015	SEC-REQ-001
FR-041	The system shall let a registered user log in with e-mail address and password and log out.	Functional	High	Team (D2)	AC-FR-041: Correct credentials open the dashboard; wrong credentials are rejected with a message that does not reveal which field was wrong; after logout, the dashboard cannot be opened without logging in again. Test: TC-015	SEC-REQ-002, SEC-REQ-006
FR-042	The system shall let a logged-in registered user create a board with a title of 1–100 characters and shall make that user the board's Owner.	Functional	High	Team (D2)	AC-FR-042: The new board appears on the creator's dashboard with role "Owner" and an empty canvas. Test: TC-016	Exactly one Owner per board
FR-043	The system shall show a logged-in registered user a dashboard listing every board they own or are a member of, with title, role and last-modified time.	Functional	Medium	Team	AC-FR-043: The dashboard lists exactly the boards where the user is Owner or member, with correct role. Test: TC-016	—
FR-044	The system shall let only the Owner delete a board after confirmation; deletion shall remove the board's content, versions, members and share links, and disconnect all participants with a notice.	Functional	High	Team (D3)	AC-FR-044: After deletion the board is gone from all dashboards, its links no longer work, connected participants see a "board deleted" notice; a delete request from a non-Owner is rejected. Test: TC-016, TC-102	FR-051
FR-045	The system shall let the Owner create a share link for a board that grants either the Viewer or the Editor role and expires after a duration chosen from 1 hour, 24 hours or 7 days (default 7 days), and shall show the Owner the list of the board's active links with role and expiry time.	Functional	High	Team (D4)	AC-FR-045: A created link appears in the list with the chosen role and correct expiry time; the full link is displayed only once, at creation. Test: TC-017	Values per D4a; SEC-REQ-004

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-046	The system shall admit a person who opens a valid share link to the board with the link's role: a logged-in registered user becomes a member with that role, unless they already have an equal or higher role; a person who is not logged in may enter a display name of 1–30 characters and join as a guest for the current browser session.	Functional	High	Team (D2, D4)	AC-FR-046: A registered user opening an Editor link appears in the members list as Editor; a guest joins with the entered name shown with "(guest)" and has the link's permissions only. Test: TC-017	D4b, D4c
FR-047	The system shall let the Owner revoke a share link; after revocation the link shall no longer admit anyone, and guests admitted by that link shall be disconnected within 5 seconds with a notice. Registered members keep their membership until the Owner removes them.	Functional	High	Team (D4)	AC-FR-047: Opening a revoked link shows "link revoked"; a guest connected through it is disconnected within 5 s; registered members admitted through it still have access. Test: TC-103	D4b; SEC-REQ-004
FR-048	The system shall reject an expired share link with an "expired" message and shall disconnect guests admitted by that link within 5 seconds after it expires.	Functional	High	Team (D4)	AC-FR-048: A link opened after its expiry time is rejected; a guest connected through it is disconnected within 5 s of expiry. Test: TC-103	D4a, D4b
FR-049	The system shall let the Owner change a member's role between Editor and Viewer or remove a member; the change shall apply to that member's next operation and to their open board view without a reload.	Functional	Medium	Team (D3)	AC-FR-049: After an Editor is changed to Viewer, their next edit is rejected and their tools are disabled; a removed member is disconnected and the board disappears from their dashboard. Test: TC-023	FR-051
FR-050	The system shall let the Owner transfer ownership of a board, after an explicit confirmation, to a registered user who is currently an Editor or Viewer member of that board; the new Owner shall gain all Owner permissions, the previous Owner shall become an Editor member, and both users' open board views shall reflect the new roles without a reload.	Functional	Medium	Team (D3)	AC-FR-050: After transfer, the new Owner can restore versions, manage links and members and delete the board, and the previous Owner can edit but can no longer do any of these; both dashboards show the new roles; a transfer requested by a non-Owner, without confirmation, to the Owner themself, to a guest or to a non-member is rejected and changes no role. Test: TC-018	Guests cannot become Owner; API in SAD §4.3 (API-02)

Req ID	Requirement (shall…)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
FR-051	The system shall permit each action only to the roles marked "Yes" in the permission matrix of §4.4 and shall reject every other attempt without changing board data or notifying other participants of the attempted change.	Functional	High	Team (D3)	AC-FR-051: For every row of the matrix, an attempt by each role marked "No" is rejected and the board data is unchanged. Test: TC-101, TC-102	SEC-REQ-003
FR-052	The system shall show every participant a list of the participants currently connected to the board, with display name and role, and shall update it when a participant joins or leaves.	Functional	Medium	Team	AC-FR-052: When a participant joins or leaves, every other participant's list changes accordingly; guests are marked "(guest)". Test: TC-017	No separate timing target in v1.0

5. Non-functional requirements (detailed)

The non-functional requirements below are measurable and tied to STP test cases. **The numeric targets in NFR-001 to NFR-005 are the specified version 1.0 targets (DECISIONS.md D7). Historical (v1.0): they had not been measured. They were measured in the execution of 12-10-2026 (uncommitted working tree on base commit 82af885; see `docs/IMPLEMENTATION_STATUS.md`) with TC-009, TC-013, TC-019 and TC-020; every target was met (values in those STP test-case records). If a measurement shows that a target cannot be met, a change request updates the requirement.**

Common measurement conditions for NFR-001 to NFR-005 (D7b):

- Server (application and MongoDB) on one laptop; clients on the same laptop; no artificial network delay; no other heavy applications running. Hardware, operating system, Node.js, MongoDB and browser versions and the Git commit are recorded with every result.
- Scripted clients run in **one Node.js process** and use one monotonic clock (`performance.now()`), so start and end times are comparable. Scripted clients use the same client synchronisation logic as the browser client (SAD COMP-02) but do not render.
- **TB-STD** (D7a) = 300 freehand strokes of at most 200 points each, 50 shapes and 20 text objects, produced by a seed script with a fixed random seed.
- Percentiles use the nearest-rank method: p95 = the value at position ⌈0.95 × n⌉ in the ascending list of n samples.
- Browser timings use the client's `board-ready` performance mark, which the client records when every object of the received state has been rendered and editing tools are enabled (SAD §4.1, test hooks).

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
NFR-001	For at least 95% of accepted board operations, the time from submission by the sending client to application by each other connected client shall be at most 500 ms.	Performance	High	Quantity: latency in ms per (operation, receiving client) pair. Start: sender calls submit. End: receiver has applied the operation to its board state. Conditions: 10 scripted clients on one board, each submitting 2 valid operations per second; 10 s warm-up excluded, then 60 s measured (≈ 1,200 operations, ≈ 10,800 samples). The sender's own acknowledgement time is recorded separately and is not part of this measure. Pass: p95 ≤ 500 ms. Test: TC-019

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
NFR-002	The system shall support at least 10 concurrently connected participants on one board while NFR-001 is met.	Capacity / Scalability	High	During the TC-019 run with 10 clients: no client disconnects; no valid operation is rejected; every accepted operation is applied exactly once by every client (no missing or duplicate sequence numbers); all final board states are identical to the server state. Pass: all four conditions hold. Test: TC-019
NFR-003	The board view shall be ready — every TB-STD object rendered and editing tools enabled — at most 3 s after navigation starts, for a logged-in Owner.	Performance	Medium	Quantity: seconds from navigation start (time origin 0) to the `board-ready` mark. Conditions: supported browser, new browser context with empty cache for every run, 10 runs. Pass: all 10 runs ≤ 3 s; all values are reported. Test: TC-020
NFR-004	After connectivity returns, a disconnected client shall show the current server state with editing tools enabled within 5 s.	Reliability	High	Quantity: seconds from the moment the client's network is re-enabled to the `board-ready` mark after resynchronisation. Conditions: one browser client offline for 30 s while another client makes 20 changes to a TB-STD board; 10 repetitions. Pass: 10/10 repetitions ≤ 5 s and the board equals the server state each time. Test: TC-009
NFR-005	A restore of a TB-STD version shall be applied by all connected participants within 3 s.	Performance	Medium	Quantity: seconds from the Owner client sending the restore request to the last of 10 connected clients applying the restored state. Conditions: 10 scripted clients (one of them the Owner) in one process; 10 repetitions. Pass: 10/10 repetitions ≤ 3 s. Test: TC-013
NFR-006	The client shall provide all core functions in the current stable desktop versions of Google Chrome, Mozilla Firefox and Microsoft Edge at a browser viewport width of 1280 px or more.	Portability / Compatibility	Medium	The core flows of TC-001, TC-002, TC-013 and TC-017 pass in each of the three browsers at a 1280 × 720 viewport with no horizontally cut-off controls; "current stable" means the release current on the test date, and versions are recorded. Test: TC-021
NFR-007	The board view shall show the user's current role and connection status at all times, show every drawing tool directly on the toolbar (no nested menus), label every toolbar control with visible text or a tooltip, and allow every toolbar control to be reached and activated with the keyboard.	Usability / Accessibility	Medium	Checklist inspection of the board view for Owner, Editor and Viewer: all four items pass for every role. Test: TC-021

The sixth target from the planning baseline — "all tested unauthorised modification attempts are rejected" — is specified as the acceptance criterion of SEC-REQ-003 (§5.1.2).

5.1 Security

5.1.1 Security objectives

Req ID	Security objective	Realised by
SEC-OBJ-001	**Confidentiality of boards:** the content, participants and version history of a board shall be accessible only to its Owner, its members and holders of a valid, unexpired, unrevoked share link.	SEC-REQ-002, SEC-REQ-003, SEC-REQ-004, SEC-REQ-007
SEC-OBJ-002	**Integrity of board content and history:** board content and version history shall change only through validated operations by roles permitted to make them, and earlier versions shall never be altered.	SEC-REQ-003, SEC-REQ-005, FR-033
SEC-OBJ-003	**Accountability and abuse resistance:** security-relevant events shall be attributable to an account or guest session whenever one has been identified (and otherwise to their request), and no single client shall be able to degrade the service for other participants by flooding requests.	SEC-REQ-006, SEC-REQ-008

5.1.2 Security requirements

Req ID	Requirement (shall…)	Type	Priority	Acceptance criteria / Test case ref
SEC-REQ-001	The system shall accept only passwords of 8 to 72 characters and shall store passwords only as salted hashes produced by an adaptive password-hashing function; passwords shall never be stored, returned or logged in plain text.	Security	High	Registration with a 7-character password is rejected; the stored user record contains a salted hash and no plain-text password; server logs contain no password. Test: TC-105
SEC-REQ-002	The system shall require a valid session token (registered users) or guest token (guests) for every request and real-time connection other than registration, login, link resolution and health check. Session tokens shall expire 24 hours after issue and become invalid at logout; guest tokens shall be valid for one board only and expire at the earlier of the link's expiry and 24 hours after issue.	Security	High	Requests and real-time connections without a token, with a tampered token, with an expired session or guest token, with a token used after logout, or with a guest token for a different board are rejected with "unauthenticated" (or "forbidden" for the other board); no board data is returned. Test: TC-106
SEC-REQ-003	The server shall check the requester's current role on the target board against the permission matrix (FR-051) for every request and every real-time message, using only server-side data to determine the role.	Security	High	**All** tested unauthorised modification attempts (100%) — through the request interface and through crafted real-time messages, including attempts that claim a different role or board in the payload — are rejected, are not stored and are not broadcast; all tested unauthorised read attempts return no board data. Test: TC-101, TC-102

Req ID	Requirement (shall…)	Type	Priority	Acceptance criteria / Test case ref
SEC-REQ-004	Share-link tokens shall contain at least 128 bits of randomness from a cryptographically secure random generator, shall be stored only in hashed form, and shall be rejected after expiry or revocation.	Security	High	Generated tokens are ≥ 22 URL-safe characters (≥ 128 bits); the database holds only token hashes; expired and revoked tokens are rejected. Test: TC-103, TC-104
SEC-REQ-005	The server shall validate every request and real-time message against a defined schema (types, required fields, value ranges, maximum text length, maximum message size of 64 KiB (65,536 bytes per Socket.IO packet)) and shall reject non-conforming input; user-supplied text shall be displayed as plain text, never interpreted as HTML or script.	Security	High	Malformed and out-of-range messages, and a message of 65,537 bytes, are rejected without changing the board; a text object containing `<script>` markup is shown literally and no script executes. Test: TC-107
SEC-REQ-006	The system shall limit failed login attempts to 5 per entered e-mail address per 15 minutes, counted in the same way whether or not an account with that address exists, and board operations to 30 per second per connection, rejecting excess requests with a "rate limited" response.	Security	Medium	The 6th failed login within 15 minutes is rejected as rate-limited even with the correct password; the same happens for an unregistered e-mail address; for both addresses every failed attempt returns the same HTTP status, error code and message (only the per-request `requestId` differs); a client sending 100 operations in one second has the operations beyond 30 rejected, while every operation of the other participants is accepted. Test: TC-108
SEC-REQ-007	In any deployed (non-local) environment, all client–server communication shall use HTTPS and secure WebSockets with TLS 1.2 or later; plain HTTP requests shall be redirected to HTTPS.	Security	High	Deployment configuration review and a connection test confirm TLS ≥ 1.2 and the redirect. The local test environment is exempt and this exemption is recorded in the STP. Test: TC-109

Req ID	Requirement (shall…)	Type	Priority	Acceptance criteria / Test case ref
SEC-REQ-008	The system shall record a security log entry for at least every authentication failure, authorisation rejection, share-link creation and revocation, member role change, ownership transfer, restore and board deletion. Each entry shall contain the event type, a UTC timestamp, the request correlation ID, the outcome and, for a rejection, a reason code; the actor (account or guest session ID) when known, otherwise the JSON value `null`; the board ID when the event concerns a board, otherwise `null`; and the internal ID of the account affected by the action (a member whose role changed or who was removed, a new Owner, or the registered account named at a failed login), otherwise `null`. Entries shall never contain passwords, session or guest tokens, share-link tokens, `Authorization` headers, the e-mail address entered at login or board content.	Security / Audit	Medium	After running the security test suite, the log contains an entry for each listed event type with every required field present; a failed login has actor `null`, board `null` and reason "unauthenticated", with the affected account ID only when the entered e-mail belongs to a registered account; a rejection carries a reason code; and a search of the log finds no password, token, entered e-mail address or board text. Test: TC-104

The values in SEC-REQ-001, SEC-REQ-002, SEC-REQ-005 and SEC-REQ-006 are recorded with their rationale in DECISIONS.md D8a–D8d. The controls that implement each security requirement are mapped in SAD §3.9.6 and the validating tests in STP §5.1.

6. Quality attributes & Acceptance tests

- **Exit criteria for acceptance:** every High-priority functional and security requirement is implemented and its test cases pass; every NFR test case passes against the targets of §5; the RTM (§8) shows every requirement linked to at least one executed test case with status "P" (passed); no open defect of severity Critical or High.
- **Acceptance test suites** (defined in STP §5 and §13):
  - Canvas editing — TC-001 to TC-005, TC-022
  - Real-time synchronisation and recovery — TC-006 to TC-011
  - Version history and recovery — TC-012 to TC-014, TC-024, TC-025
  - Accounts, boards, sharing and roles — TC-015 to TC-018, TC-023
  - Performance and capacity — TC-019, TC-020 (and the timing parts of TC-009, TC-013)
  - Compatibility and usability — TC-021
  - Security — TC-101 to TC-109

- **Quality attributes and their measures:** responsiveness (NFR-001, NFR-003, NFR-005), capacity (NFR-002), reliability and recoverability (FR-025, FR-027, NFR-004), consistency (FR-021 to FR-023), usability (NFR-007), portability (NFR-006), security (§5.1).
- **Status of results:** historical (v1.0): no test had been executed. After the execution of 12-10-2026 (uncommitted working tree on base commit 82af885; see `docs/IMPLEMENTATION_STATUS.md`): 32 test cases Pass, 2 Blocked (TC-021, TC-109), none Fail; RTM status P for every requirement except NFR-006, NFR-007 and SEC-REQ-007 (N). Addendum 17-10-2026: TC-021 was executed manually and set to Pass with limitations (STP TC-021 record), so 33 test cases are Pass and 1 Blocked (TC-109), none Fail; RTM status P for every requirement except SEC-REQ-007 (N). The STP exit criteria are not yet met (TC-109 Blocked; four-member review of the test summary outstanding, STP §14). Results will be recorded in the STP test-case "Actual result" and "Status" fields, and the RTM status is derived from them by the rules in §8. Addendum 02-11-2026: the final regression of `main` after PR #33 (`docs/reports/RELEASE_READINESS_2026-11-02.md`; 210 unit and integration tests, 31 E2E tests in Chromium, Chrome and Edge and in Playwright's Firefox and WebKit builds) found no failure; test-case statuses and RTM statuses are unchanged (33 Pass, 1 Blocked: TC-109).

7. System models and diagrams

The template requires at least two UML use-case diagrams. Diagram sources are in `docs/diagrams/src/` (PlantUML) and rendered images in `docs/diagrams/png/`.

7.1 UML Use-Case diagram UC-01 — Collaborate on a board

Figure 1 shows the use cases for working on a board.

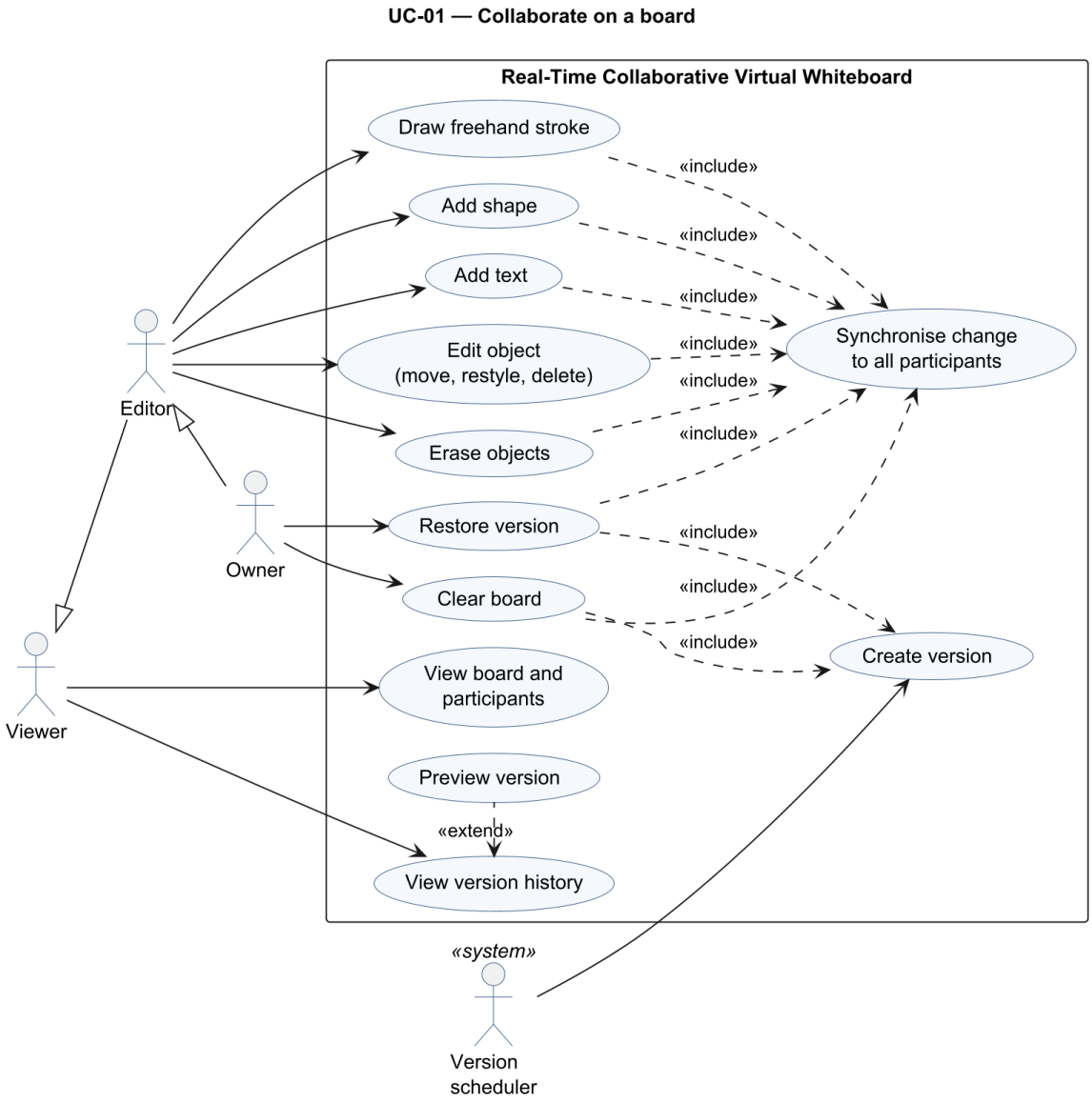


Figure 1 — UC-01 Collaborate on a board

Actors: **Viewer**, **Editor** (inherits Viewer), **Owner** (inherits Editor), and the **Version scheduler** (a time-based system actor that triggers automatic versions).

Use case	Primary actor	Requirements
View board and participants	Viewer	FR-024, FR-052
View version history / Preview version	Viewer	FR-031, FR-032
Draw freehand stroke	Editor	FR-001, FR-002, FR-008
Add shape / Add text	Editor	FR-003, FR-004, FR-008
Edit object (move, restyle, delete)	Editor	FR-005

Use case	Primary actor	Requirements
Erase objects	Editor	FR-006
Clear board («include» Create version — creates a version only if the content changed, §4.3)	Owner	FR-007, FR-030
Restore version («include» Create version — always creates a version)	Owner	FR-033, FR-034
Create version automatically (every *T* minutes, subject to the no-change rule)	Version scheduler	FR-030
Synchronise changes («include» from every editing use case)	—	FR-020 to FR-023

7.2 UML Use-Case diagram UC-02 — Access and sharing

Figure 2 shows the use cases for accounts, boards and sharing.

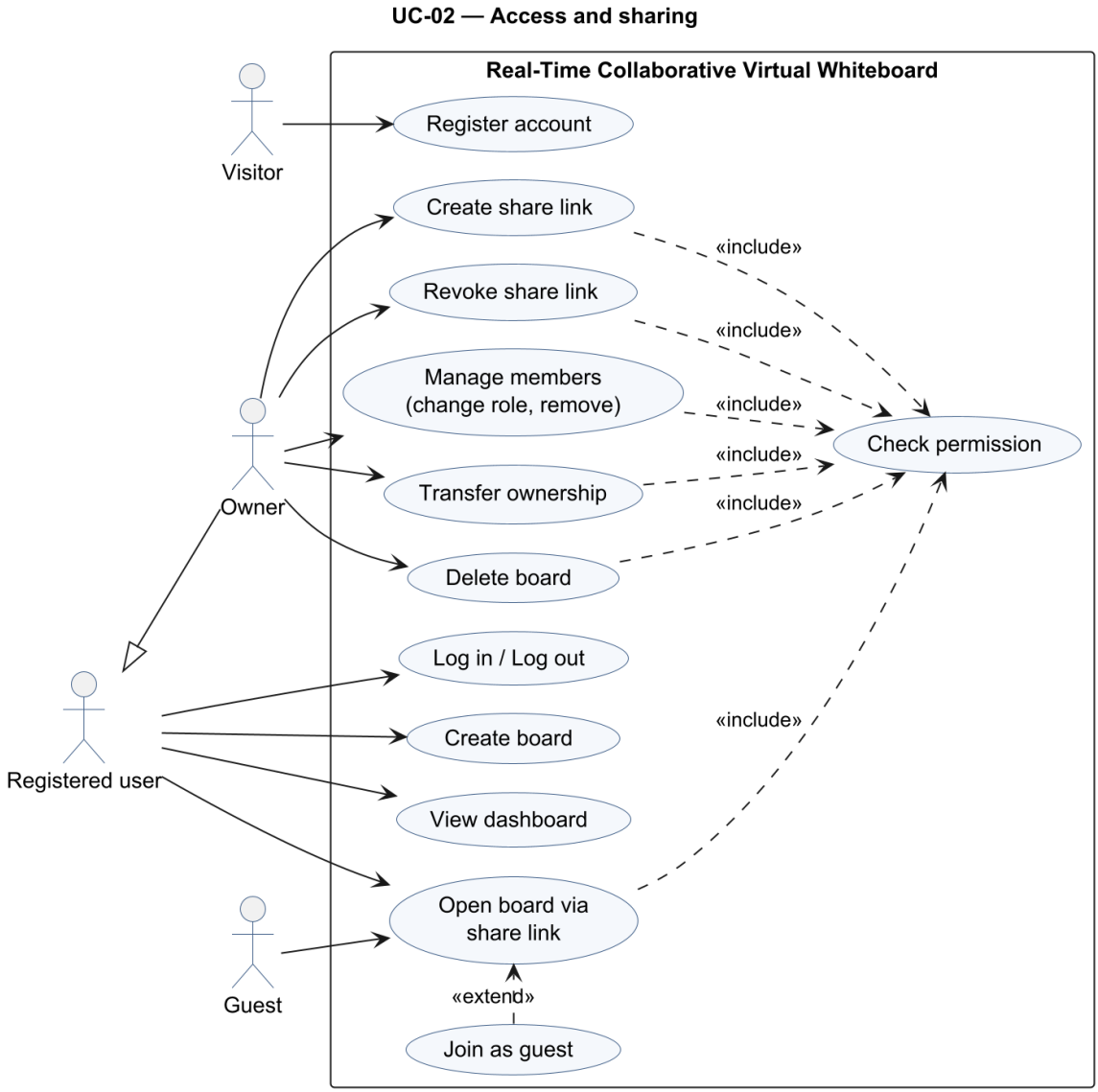


Figure 2 — UC-02 Access and sharing

Actors: **Visitor**, **Registered user**, **Guest**, **Owner** (inherits Registered user).

Use case	Primary actor	Requirements
Register account	Visitor	FR-040
Log in / Log out	Registered user	FR-041
Create board / View dashboard	Registered user	FR-042, FR-043
Open board via share link	Registered user, Guest	FR-046
Join as guest («extend» Open board via share link)	Guest	FR-046
Create share link / Revoke share link	Owner	FR-045, FR-047
Manage members (change role, remove)	Owner	FR-049
Transfer ownership (to an Editor or Viewer member)	Owner	FR-050
Delete board	Owner	FR-044
Check permission («include» from every protected use case)	—	FR-051, SEC-REQ-003

8. Requirements Traceability Matrix (RTM)

Status codes (derived from the STP test-case status, STP §13): **P** = every linked test case has status Pass; **A** = at least one linked test case has status Fail (attempted, defect open); **N** = otherwise — at least one linked test case is Not executed or Blocked and none has failed (a Blocked test is named in the Comments column with its reason). Historical (v1.0): all entries were **N** because no test had been executed. The statuses below are derived from the execution of 12-10-2026 (uncommitted working tree on base commit 82af885; see `docs/IMPLEMENTATION_STATUS.md`). Update 17-10-2026: the table below is the current master RTM (`docs/traceability/rtm.md`); NFR-006 and NFR-007 reflect TC-021 Pass with limitations of 17-10-2026. The columns follow the SRS template: "Section ref / Design Spec" gives the SRS section and the SAD sequence-diagram (SD) and API identifiers; "Module" gives the SAD component (COMP) identifiers.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/ P/A)	Comments
FR-001	Freehand stroke	SRS 4.1 / SD-01, API-01	COMP-01, COMP-02, COMP-06	TC-001	P	
FR-002	Colour and stroke width	SRS 4.1 / API-01	COMP-01	TC-001	P	
FR-003	Rectangle, ellipse, line	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-002	P	
FR-004	Text objects	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-002	P	
FR-005	Select, move, restyle, delete	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-003	P	
FR-006	Object eraser	SRS 4.1 / API-01	COMP-01, COMP-02, COMP-06	TC-004	P	
FR-007	Owner-only clear; pre-clear content kept; recovery	SRS 4.1 / SD-05, API-01	COMP-05, COMP-06, COMP-07, COMP-08	TC-005, TC-025	P	
FR-008	Board content size limit 15 MiB (M3)	SRS 4.1 / SD-01, API-01	COMP-01, COMP-06	TC-022	P	
FR-020	Broadcast accepted operations	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-001, TC-002	P	
FR-021	Server sequence numbers and ordering	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-006, TC-007	P	
FR-022	Independent edits preserved	SRS 4.2 / SD-01	COMP-06	TC-006	P	
FR-023	Same-object last write wins	SRS 4.2 / SD-01	COMP-02, COMP-06	TC-007	P	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/ P/A)	Comments
FR-024	Full state on join	SRS 4.2 / SD-03, API-01	COMP-06, COMP-08	TC-008	P	
FR-025	Disconnect indicator and resync	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-009	P	
FR-026	Rejection feedback and rollback	SRS 4.2 / SD-01, API-01	COMP-02, COMP-06	TC-010	P	
FR-027	Persist before broadcast	SRS 4.2 / SD-01	COMP-06, COMP-08	TC-011	P	
FR-030	Automatic versions, no-change rule	SRS 4.3 / API-03	COMP-07, COMP-08	TC-005, TC-012	P	
FR-031	Version list for all roles	SRS 4.3 / API-03	COMP-01, COMP-07	TC-012	P	
FR-032	Read-only version preview	SRS 4.3 / API-03	COMP-01, COMP-07	TC-012	P	
FR-033	Owner-only restore as new version; recovery	SRS 4.3 / SD-02, API-03	COMP-05, COMP-07, COMP-08	TC-013, TC-024, TC-102	P	
FR-034	Restored state pushed to all	SRS 4.3 / SD-02, API-01	COMP-06, COMP-07	TC-013	P	
FR-035	Retain at least 20 versions	SRS 4.3 / API-03	COMP-07, COMP-08	TC-014	P	
FR-040	Register account	SRS 4.4 / API-02	COMP-03, COMP-04	TC-015	P	
FR-041	Log in and log out	SRS 4.4 / API-02	COMP-03, COMP-04	TC-015	P	
FR-042	Create board, creator is Owner	SRS 4.4 / API-02	COMP-03, COMP-05	TC-016	P	
FR-043	Dashboard of boards	SRS 4.4 / API-02	COMP-01, COMP-03, COMP-05	TC-016	P	
FR-044	Owner-only board deletion	SRS 4.4 / API-02	COMP-05, COMP-06	TC-016, TC-102	P	
FR-045	Create share link with role and expiry	SRS 4.4 / API-02	COMP-05	TC-017	P	
FR-046	Join via link (member or guest)	SRS 4.4 / SD-03, API-02	COMP-04, COMP-05	TC-017	P	
FR-047	Revoke share link	SRS 4.4 / SD-03, API-02	COMP-05, COMP-06	TC-103	P	
FR-048	Expired link rejected	SRS 4.4 / SD-03	COMP-05, COMP-06	TC-103	P	
FR-049	Change member role or remove	SRS 4.4 / API-02	COMP-05, COMP-06	TC-023	P	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/ P/A)	Comments
FR-050	Transfer ownership to a member	SRS 4.4 / SD-04, API-02	COMP-03, COMP-05, COMP-06	TC-018	P	
FR-051	Permission matrix enforced	SRS 4.4 / SD-01, SD-02, SD-03, SD-04, SD-05	COMP-03, COMP-05, COMP-06	TC-101, TC-102	P	
FR-052	Participant presence list	SRS 4.4 / API-01	COMP-01, COMP-06	TC-017	P	
NFR-001	95% of updates visible within 500 ms	SRS 5 / SD-01	COMP-02, COMP-06, COMP-09	TC-019	P	
NFR-002	At least 10 concurrent participants	SRS 5	COMP-06	TC-019	P	
NFR-003	TB-STD loads within 3 s	SRS 5	COMP-01, COMP-03	TC-020	P	
NFR-004	Reconnect recovery within 5 s	SRS 5 / SD-01	COMP-02, COMP-06	TC-009	P	
NFR-005	Restore visible to all within 3 s	SRS 5 / SD-02	COMP-06, COMP-07	TC-013	P	
NFR-006	Chrome, Firefox, Edge support	SRS 5	COMP-01	TC-021	P	TC-021 Pass with limitations (17-10-2026): manual in Firefox 158 and Chrome, automated in Chrome and Edge; Edge not checked by hand; the automated run predates the stroke-width tooltip fix
NFR-007	Role/status visible, labelled toolbar	SRS 5	COMP-01	TC-021	P	TC-021 Pass with limitations (17-10-2026): stroke-width tooltip defect found and fixed during TC-021, re-checked in Firefox only
SEC-REQ-001	Password policy and salted hashing	SRS 5.1.2 / API-02	COMP-04, COMP-08	TC-105	P	
SEC-REQ-002	Tokens required, expiry, logout	SRS 5.1.2 / SD-03, API-01, API-02	COMP-03, COMP-04, COMP-06	TC-106	P	
SEC-REQ-003	Server-side authorisation of every request/event	SRS 5.1.2 / SD-01, SD-02, SD-03, SD-04, SD-05	COMP-03, COMP-05, COMP-06	TC-101, TC-102	P	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SEC-REQ-004	Link token entropy, hashing, expiry	SRS 5.1.2 / SD-03, API-02	COMP-05, COMP-08	TC-103, TC-104	P	
SEC-REQ-005	Schema validation, size limit, no HTML	SRS 5.1.2 / API-01, API-02, API-03	COMP-01, COMP-03, COMP-06	TC-107	P	
SEC-REQ-006	Login and operation rate limits	SRS 5.1.2 / API-01, API-02	COMP-03, COMP-06	TC-108	P	
SEC-REQ-007	TLS 1.2+ in deployment	SRS 5.1.2	COMP-03, COMP-06	TC-109	N	TC-109 Blocked: no deployed HTTPS instance
SEC-REQ-008	Security event logging without secrets	SRS 5.1.2 / SD-04	COMP-03, COMP-04, COMP-05, COMP-06, COMP-07, COMP-09	TC-104	P	

Security objectives are traced through the security requirements that realise them (SRS 5.1.1):

Objective	Realised by	Test case(s)	Status (N/P/A)
SEC-OBJ-001	SEC-REQ-002, SEC-REQ-003, SEC-REQ-004, SEC-REQ-007	TC-101, TC-102, TC-103, TC-104, TC-106, TC-109	N
SEC-OBJ-002	SEC-REQ-003, SEC-REQ-005, FR-033	TC-013, TC-101, TC-102, TC-107	P
SEC-OBJ-003	SEC-REQ-006, SEC-REQ-008	TC-104, TC-108	P